FHV
Vorarlberg University
of Applied Sciences

State Machines
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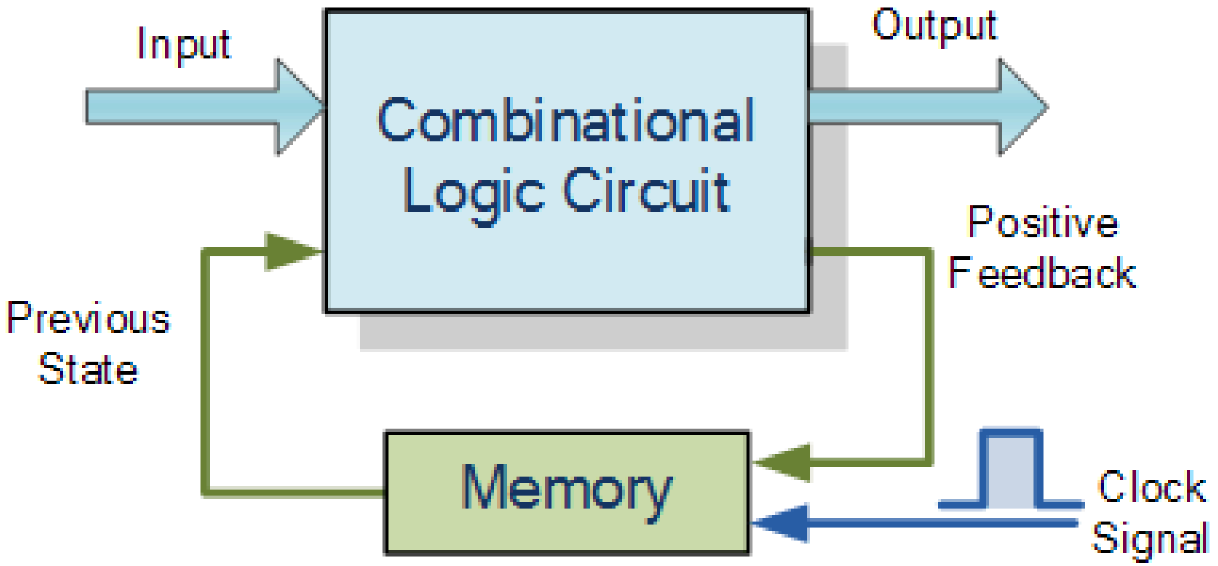
Sequential Logic

- In sequential logic the outputs depends on the present inputs and on the history of the inputs
- Sequential logic has memory



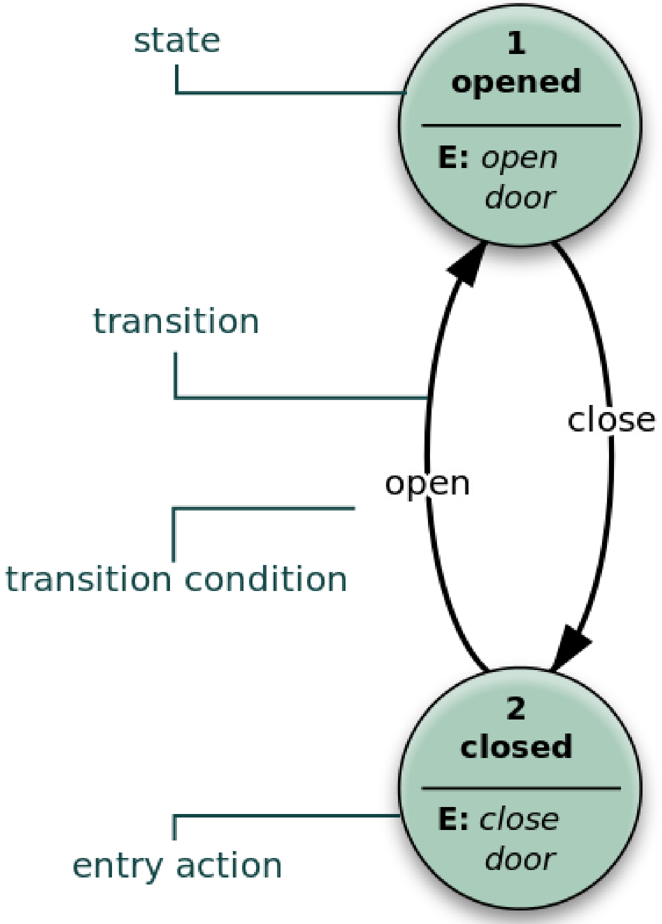
Finite State Machine

An example finite state machine
[(endlicher) (Zustands-)Automat]
including the basic elements:

- States
- Transitions
- Conditions
- Actions

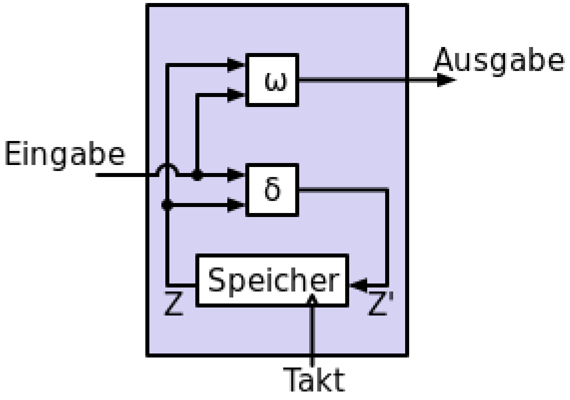
Can be implemented with:

- PLCs
- Software
- Hardware

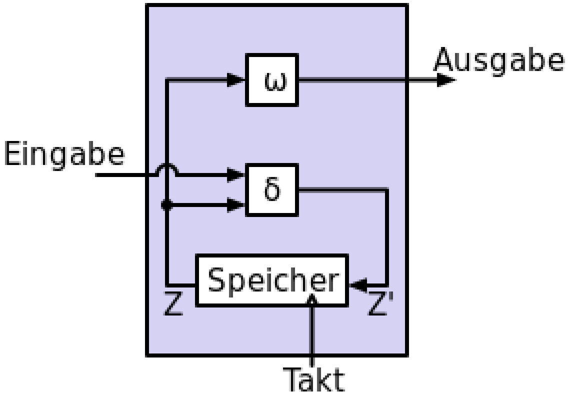


Mealy and Moore State Machines

- Moore: outputs depend only of states
- Mealy: outputs depend of states and inputs
- Mealy: less states, reacts faster on inputs
- Moore: safer (synchronous)

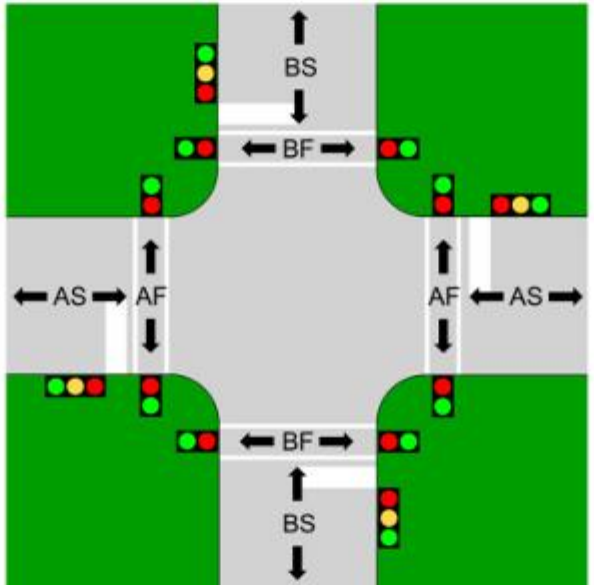


Mealy Automat



Moore Automat

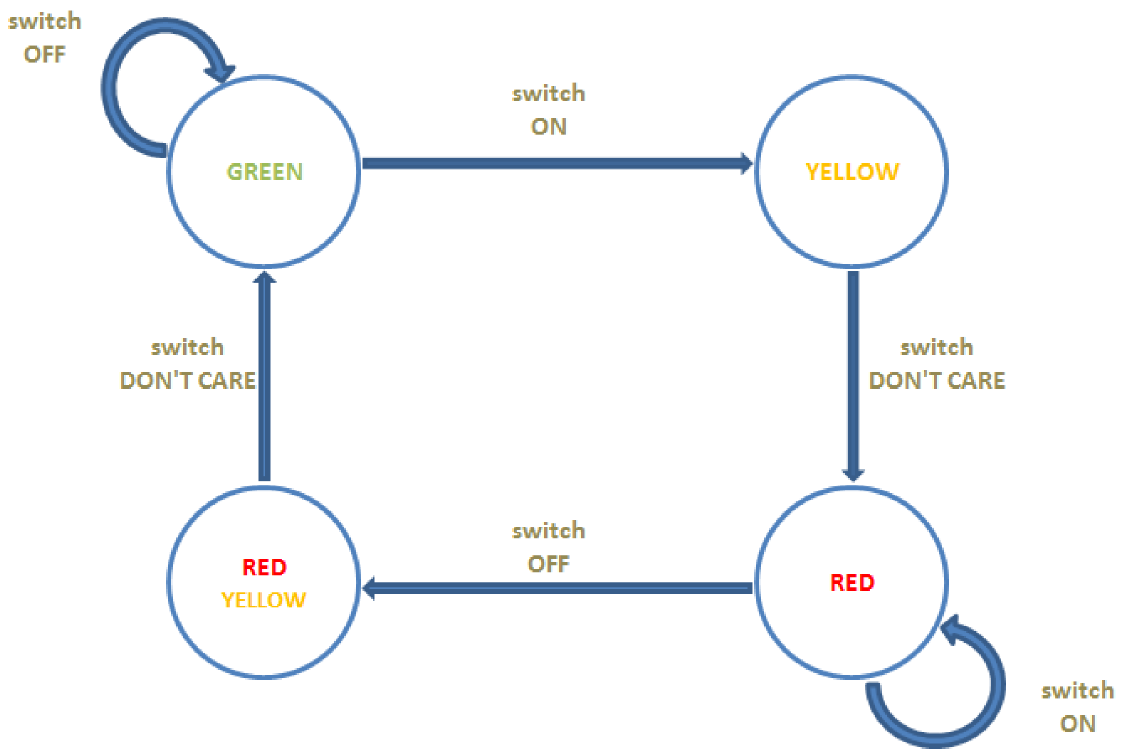
Traffic Light Control I



Assumptions

- only one pedestrian traffic light is in the focus
- Switch will start switching process – starting at “green” – ending at “red”
- Switch back will start process again – ending at “green”

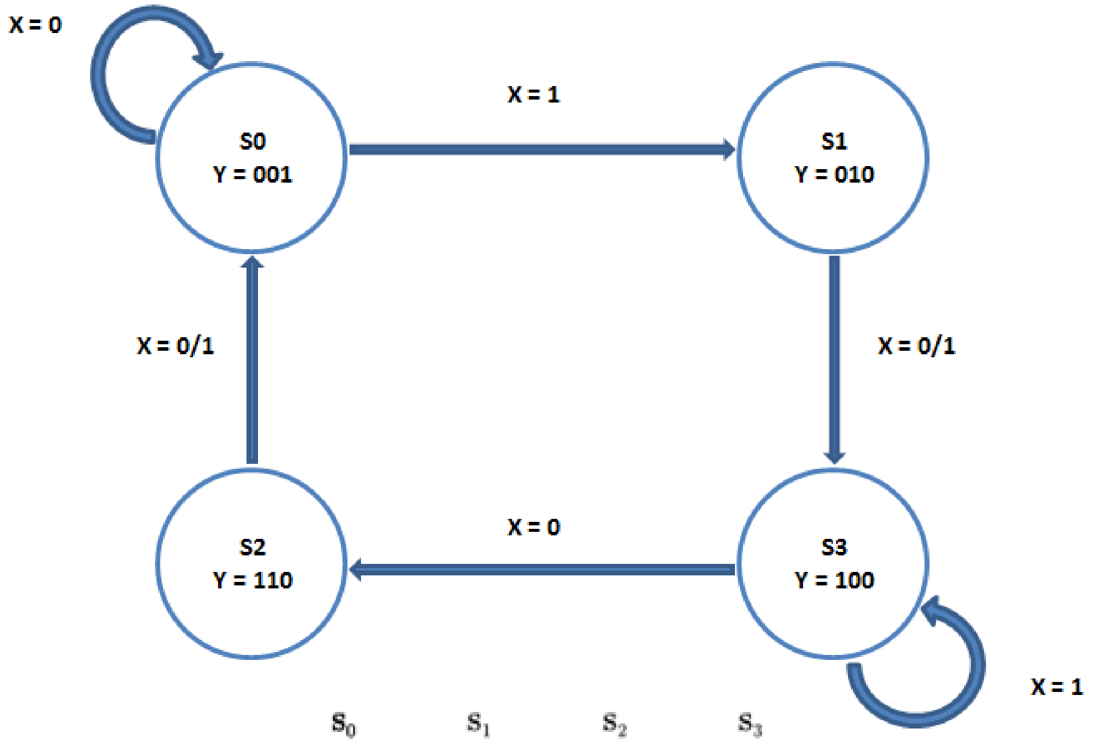
Traffic Light Control II

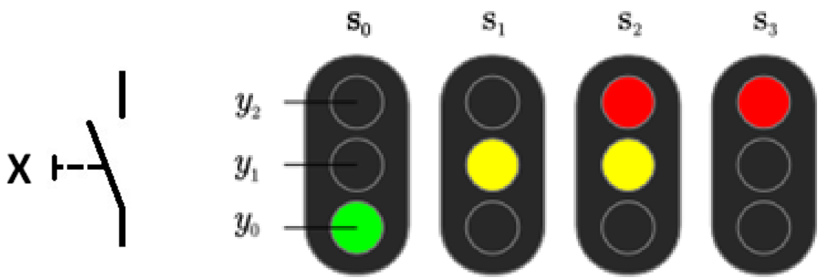


State Machine – Moore type

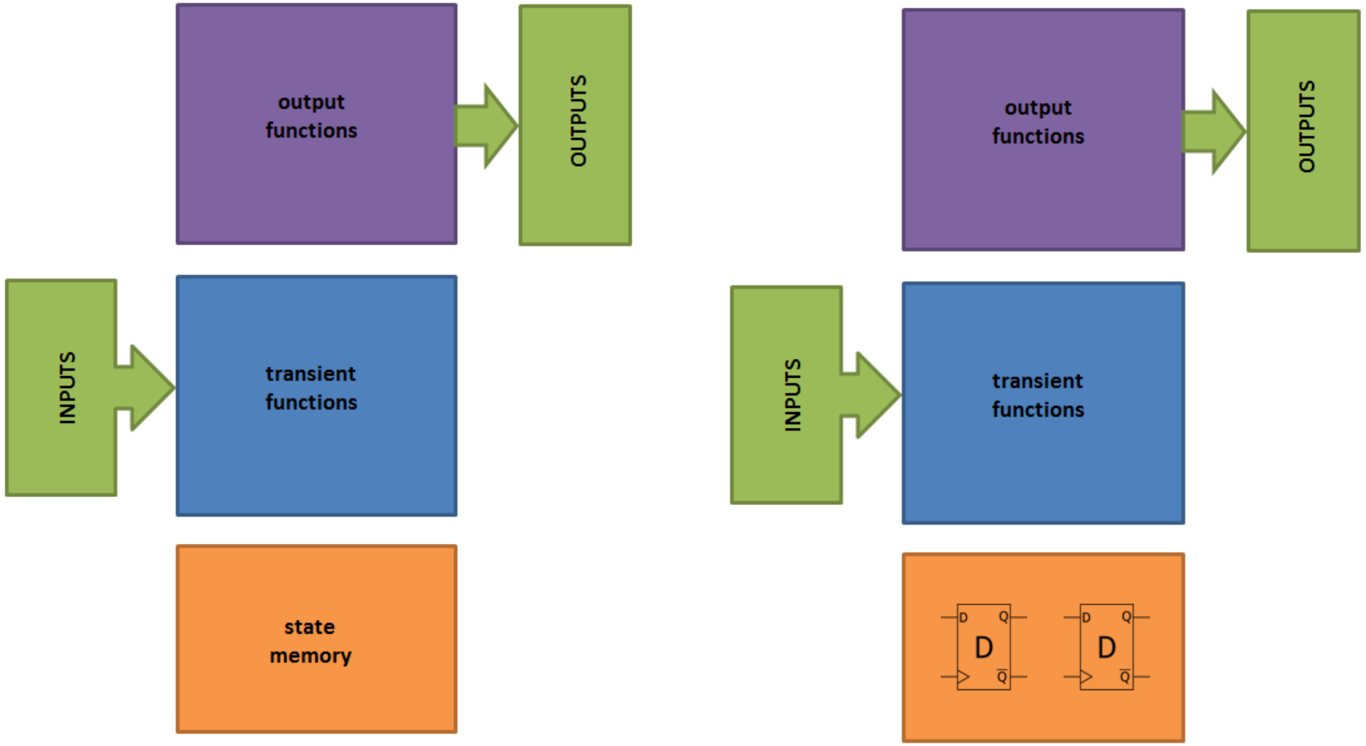
- Outputs only depend on the actual state

Traffic Light Control III





Traffic Light Control IV



Moore machine

- The four states are represented by two D-type flip-flops
- Output and transient functions can be analyzed independently

Traffic Light Control V

- Transient functions

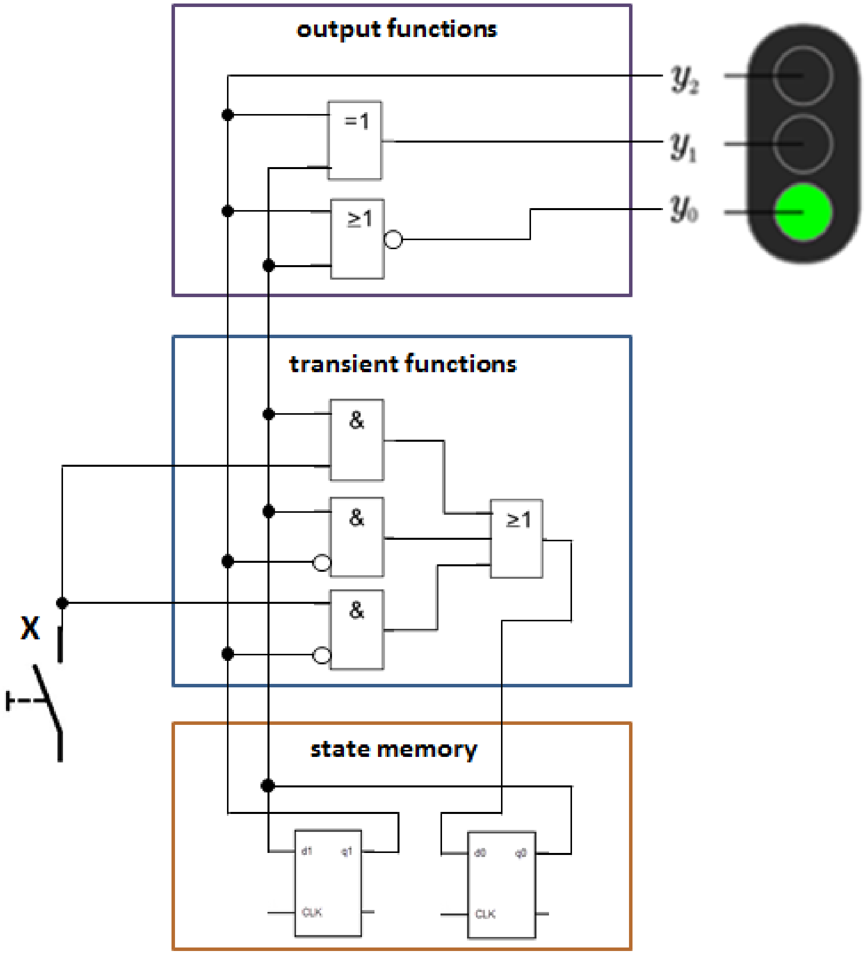
state	q_1	q_0	x	d_1	d_0
s_0	0	0	0	**0**	**0**
s_0	0	0	1	**0**	**1**
s_1	0	1	0	**1**	**1**
s_1	0	1	1	**1**	**1**
s_2	1	0	0	**0**	**0**
s_2	1	0	1	**0**	**0**
s_3	1	1	0	**1**	**0**
s_3	1	1	1	**1**	**1**

Traffic Light Control VI

- Output functions

state	q_1	q_0	y_2	y_1	y_0
s_0	0	0	**0**	**0**	**1**
s_1	0	1	**0**	**1**	**0**
s_2	1	0	**1**	**1**	**0**
s_3	1	1	**1**	**0**	**0**

Traffic Light Control VII



Garage Door Control I

For a garage door a state machine has to be implemented.

The state diagram and the transition and output functions must be implemented.

Main task:

- A push button is used both to open and close the door and triggers the "T" signal.
- Two stop sensors give signals ("O" or "U"),when the door is either fully open or fully closed.
- The state machine shall generate the signals "M“ and "R", where M indicates whether the motor is activated (= 1). When the motor is activated, R determines the direction of movement of the motor (1 = up, 0 = down).
- If the door is closed, the opening process is started; if the door is open, the closing process is started.
- As soon as the (closing) door is closed, the motor is switched off;
  similarly, the motor is switched off as soon as the (opening) door is open.

Garage Door Control II

Expand the state machine as follows:

- A sensor is also attached to the door, which interrupts the closing process on contact and switches to the opening process. This generates the "K" signal. Please note that the contact sensor should have priority over the stop sensor.
- If the push button is activated during a closing or opening process, the door should stop. It should then move in the opposite direction when the button is pressed again.

Also consider the following problem without implementing it:

- The state machine shall monitor the hardware for functionality as good as possible.
  Which additional sensors are required?
  How does the state diagram (and the functions) change?

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